

Total No. of Questions :6]

SEAT No. :

P3663

[Total No. of Pages : 2

APR - 15/ENGG. - 148

T.E. (Information Technology) (In Sem - Semester - II)

OPERATING SYSTEM

(2012 Pattern)

Time :1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) *Answer Q.1. or Q.2, Q.3. or Q.4, Q.5. or Q.6.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Explain the four basic functions of Operating Systems? [6]

b) Explain micro-kernel architecture with neat diagram. [4]

OR

Q2) a) What is a virtual machine? Explain the concept of virtualization. [6]

b) Enlist any four types of Operating Systems and give an example of each.[4]

Q3) a) Suppose that a process spawns another process using fork system call. What if the child process completes the execution before parent process? What if the parent process completes the execution before child process? How a parent process realizes the termination of a child process? Who is the ancestor of processes under UNIX? Suppose child process loads another binary program using execve system call and the original program is to execute the code written after execve statement, would that be executed? [6]

b) Draw and explain the five state process state diagram. [4]

P.T.O.

OR

- Q4) a)** For the table given below, calculate average waiting time and average turnaround time and draw a Gantt chart illustrating the process execution using Round Robin scheduling algorithm. A larger number indicates higher priority. [6]

Process	Arrival Time	Burst Time	Priority
A	0	3	5
B	2	6	2
C	4	4	4
D	6	5	3
E	8	2	1

- b) Differentiate between user-level threads and kernel-level threads. [4]

- Q5) a)** Enlist different ways to handle mutual exclusion. Give an example of each. [4]

- b) What is a counting semaphore? Explain with pseudo-code where you do apply this synchronization construct? [6]

OR

- Q6) a)** The following codes are executed concurrently on the uniprocessor system. What could be the problem with the following code? If there is a problem, explain what it is. [4]

```
void procl ( ) {
    pthread_mutex_lock (&m1);
    /* use object 1 */
    pthread_mutex_lock(&m2);
    /* use objects 1 and 2 */
    pthread_mutex_unlock (&m2);
    pthread_mutex_unlock (&m1);
}

void proc2 ( ) {
    pthread_mutex_lock(&m2);
    /* use object 2 */
    pthread_mutex_lock (&m1);
    /* use objects 1 and 2 */
    pthread_mutex_unlock (&m1);
    pthread_mutex_unlock (&m2);
}
```

- b) Explain the necessary and sufficient conditions for the occurrence of a deadlock. [6]

